



# Designing a Machine Learning Search Engine for Nuclear Safety Record Searches

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*MSc. Dissertation: School of Computer Science and Engineering, Bangor University*

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# Project Aims & Objectives



## PROJECT AIM

To develop a tool that retrieves relevant nuclear incident reports from natural-language queries and surfaces recurring failure patterns, making decades of nuclear safety records genuinely searchable by the public and analysts alike.

### Why it matters

Public nuclear safety data is vast, technical, and fragmented across agencies: a plain-English search closes the gap between raw records and public understanding.



### Curate the corpus

Collect nuclear incident report datasets from public regulatory sources (IAEA, NRC, INES).



### Baseline lexical search

Build a TF-IDF and cosine-similarity retrieval system as the performance floor.



### Semantic search layer

Implement Sentence-BERT embeddings to understand plain-English queries.



### Failure-pattern clustering

Apply clustering techniques to reveal recurring incident themes.



### Public-facing interface

Develop an accessible, web-based UI using Streamlit.



### Rigorous evaluation

Measure search accuracy, relevance, and usability against baselines.

# Progress So Far

*Foundational research phase*



## Literature Review

- Surveyed existing search and information-retrieval systems for safety-critical, technical document collections.
- Reviewed NLP approaches used in prior nuclear and industrial incident analysis tools.
- Compared lexical (TF-IDF) vs. semantic (transformer-based) retrieval methods for domain suitability.



## Research Design & Plan

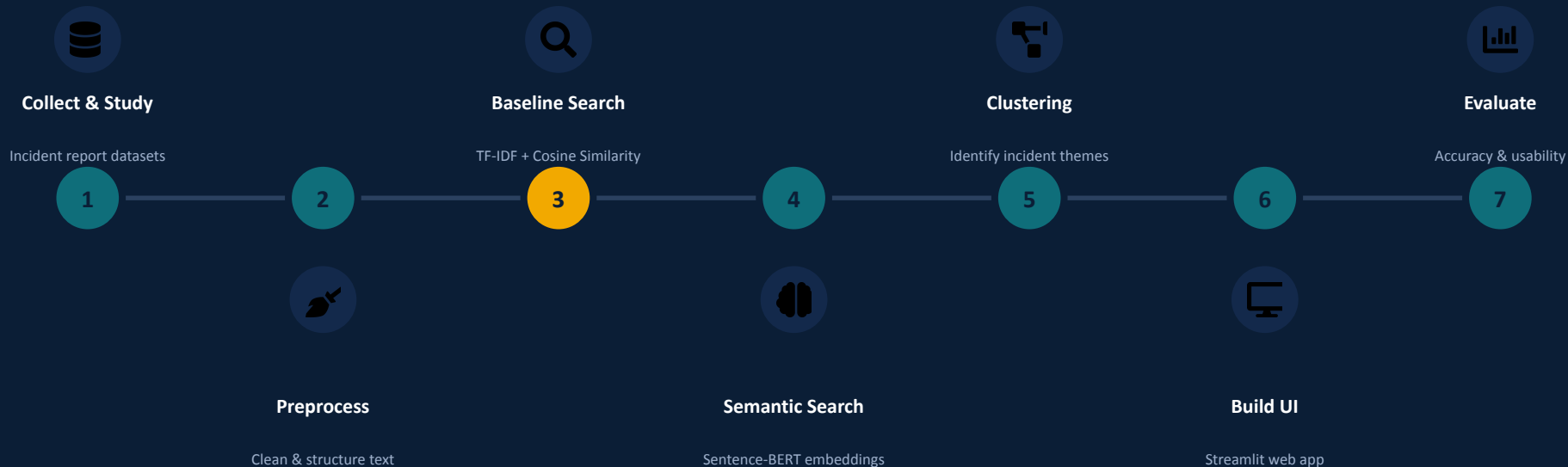
- Defined the end-to-end system architecture: ingestion → preprocessing → retrieval → clustering → interface.
- Selected the technology stack: Python, Pandas, NLTK, Scikit-learn, Sentence-BERT, Streamlit.
- Drafted an evaluation methodology (precision/recall, usability testing) and a staged implementation timeline.



**Where this leaves us:** a validated problem framing, an evidence-based architecture, and a concrete implementation plan, ready to move from design into build.

# What's Ahead

*Implementation roadmap: from raw records to a working search engine*



**Next milestone:** a design diagram of a working TF-IDF baseline retrieval system, benchmarked against the semantic search approach.